

English Retention after Tagalog Continuation

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Abstract

Pajo (2026) trained nanochat from scratch on WikiText-TL-39 and measured held-out Tagalog bits-per-byte (BPB) under equal exposure (AsPredicted #306780). That study is complete and is not amended here. Those Tagalog-only checkpoints never saw English, so they cannot answer whether a model “forgot English because it learned Tagalog.” This Stage 1 manuscript locks the complementary experiment, still entirely inside Karpathy nanochat: `tok_train` `base_train` `official` `evaluate_bpb`, commit 92d63d4, depth dial only. Pretrain on WikiText-103 raw (English-train 32,768 BPE, `official`), prove provenance, then continue on the frozen P1.1 WikiText-TL-39 train split. Dual confirmatory benchmarks: English `val_bpb_full` (retention / forgetting) and Tagalog `val_bpb_full` on `reconstructed_article_70_15_15` (Filipino acquisition). The forgetting contrast is extra-English continuation versus Tagalog continuation at the same token budget. P1.1 Table 2 is the Tagalog from-scratch ceiling and the English negative control. New GitHub and Hub repositories; P1.1 weights stay read-only. No phase-2 numbers are invented. No Hugging Face Trainer, CORE, chat, or SFT in the primary table.

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Keywords: catastrophic forgetting; English retention; Tagalog acquisition; WikiText-103; WikiText-TL-39; bits-per-byte; provenance; equal-exposure; nanochat; preregistration.

arXiv categories: cs.CL, cs.LG.

Status: Stage 1. Direction recalibrated 16 August 2026. English parents and Tagalog continuation have not been run.

P1.1 (frozen, A/P/C/L/N):

https://www.researchgate.net/publication/412302216_Equal-Exposure_Depth_and_Held-Out_Tagalog_Bits-per-Byte_on_WikiText-TL-39; code

<https://github.com/pageman/nanochat-filipino>; weights

<https://huggingface.co/pageman/nanochat-filipino-p1-fixed-d20-3x>.

This study (new repos, not written onto P1.1): `pageman/nanochat-filipino-p2-en-then-tl` on GitHub and Hugging Face.

Direction lock: `docs/papers/p2-cf-english/DIRECTION-RECALIBRATION.md`.

What changed, and why P1.1 cannot be the CF parent

The question is: after English pretraining, how much English is lost when the same decoder is continued on Filipino Wikipedia text, and did it actually learn Filipino?

WikiText-TL-39 is a corpus, not a model that was “pretrained in English.” English pretraining is a separate nanochat run on WikiText-103 raw . Filipino tuning uses the frozen P1.1 train split of WikiText-TL-39 .

P1.1 Hub checkpoints are a pure Tagalog speedrun . Using them as the CF parent would measure Tagalog retention under English data — the opposite causal arrow. That draft is retired.

P1.1 remains the measurement instrument for Filipino and the negative control for English (a model that was *not* English-pretrained).

Glossary

Frozen Tagalog-from-scratch study. Not the English parent.

English-from-scratch nanochat on WikiText-103 raw. This *is* the CF parent.

Provenance battery, scored before any Tagalog training token.

Frozen EN0, dual eval (English before, Tagalog before).

Extra English continuation, (drift).

Tagalog continuation on P1.1 train, (CF intervention).

50/50 English/Tagalog documents, .

English `val_bpb_full(A2)` `val_bpb_full(A1)`. Positive: Tagalog data hurt English more than more English did.

Tagalog `val_bpb_full(A2)` versus P1.1 Table 2 (from-scratch ceiling) and versus A0 (English parent before Filipino).

After A2, English `val BPB` EN0 untrained, or versus A0, or English `val BPB` no longer beats the English byte unigram. Not a bump.

Bits per UTF-8 byte. Fair across the English BPE and the P1.1 Tagalog BPE. Do not mix tokenizers in one training run.

P1.1 facts used as references, not as parents

Tagalog `val_bpb_full` seed 0: d8 , d12 , d16 , d20
. Untrained ; unigram . Split
`reconstructed_article_70_15_15`. nanochat 92d63d4. . One Tagalog test
BPB on d20; this study does not reuse it as a post-continuation number.

Confirmatory question

File a new AsPredicted before EN0 confirmatory BPB (pilots allowed). Then:

When an English-pretrained nanochat (EN0), whose English provenance has passed P0, is continued for tokens on WikiText-TL-39 train, does English held-out BPB rise more than under matched extra-English continuation, and does Tagalog held-out BPB fall toward the P1.1 from-scratch ceiling?

Prediction: by at least BPB at the registered depths. Tagalog A2 BPB will be strictly below A0 Tagalog BPB (learning Filipino) but need not match P1.1 d20 (different tokenizer, different init). Depth rankings of require the rule and more than one seed. P0 failure aborts A2.

Provenance battery P0

“Was it really pretrained in English?” is an empirical gate, not a README claim.

On a frozen English validation split, before any Tagalog train token:

1. EN0 English val_bpb_full same-depth untrained.
2. EN0 English val_bpb_full P1.1 d20 English val_bpb_full on the same UTF-8 (P1.1 Tagalog BPE), by BPB.
3. EN0 Tagalog val_bpb_full P1.1 d20 Tagalog val_bpb_full (has not learned Filipino yet), by BPB.
4. English-train BPE hashes recorded; train log names WikiText-103 only; no ClimbMix.

If any check fails, EN0 is not an English parent.

Related work

Catastrophic interference is sequential overwriting of old knowledge . Typical LLM CF starts from an English (or multilingual) base and adds a new task . This paper stays in that causal direction, but keeps P1.1's purity: WikiText-family corpora, nanochat depth dial, BPB rather than CORE . EWC and replay are later . Equal-exposure arithmetic is copied from P1.1 so English pretraining is not a silent compute confound .

Methods

Lineage

New AsPredicted. New GitHub/Hub pageman/nanochat-filipino-p2-en-then-tl. P1.1 repos read-only. Authority: new PDF DIRECTION-RECALIBRATION.md this manuscript ledger deviation.

nanochat-only contract

Every trained weight in this study is a nanochat decoder from commit 92d63d4e8bb4df75c3b71618f31ddde2378b2bcd, plus the documented NANOCHAT_DATA_DIR hook and no other model-file edits. English BPE: python -m scripts.tok_train on WikiText-103 raw train. English pretrain, extra-English, Tagalog continuation, and joint mix: python -m scripts.base_train with --depth, --max-seq-len=2048, --core-metric-every=-1. Both language scores: official evaluate_bpb through a copy of scripts/p1/gate_j_full_bpb.py. P1.1 d20 used as a negative control is the already-published nanochat checkpoint, not a retrained Hugging Face model. Forbidden: Hugging Face Trainer or from_pretrained as the language model, llama.cpp, python -m nanochat.dataset, CORE, SFT, a second architecture, or loading P1.1 model_000294.pt as the English parent.

English phase (EN0)

Corpus: WikiText-103 *raw*. LF-only. Article-hash 70/15/15. Tokenizer: 32,768 BPE on English *train* only. Trainer: nanochat 92d63d4, , from English with explicit `--num-iterations`, ratio never , CORE off. Depths: 8 and 20 required; 12 and 16 if budget. Seed 0 confirmatory for EN0; extra seeds later.

Tagalog phase (A2)

Data: frozen P1.1 train JSONL only (hash 2b0474c5...). Encode with the *English* BPE. English-BPE tokens unless the PDF names English . Fresh optimizer, peak LR EN0 peak, warmup 14. Do not load P1.1 model_000294.pt as the start weight.

Arms

Arm	Start	Phase-2 stream
EN0	random	English train (pretrain)
A0	EN0	none (dual eval)
A1	EN0	more English
A2	EN0	P1.1 Tagalog train
A3	EN0	50/50 documents

Dual evaluators

Same `evaluate_bpb` wrapper as P1.1, two data roots. English val never in Tagalog paths. Seal English and Tagalog acquisition on validation with both tests unread. Then at most one English test and at most one Tagalog test, predeclared. Loop `eval-tokens=262144` is diagnostic.

What is not done

No ClimbMix. No `python -m nanochat.dataset`. No P1.1 weight as English parent. No overwrite of `p1-fixed-d20-3x`. No Hugging Face Trainer. No confirmatory SFT/CORE/EWC. No invented rater scores. No mixing tokenizers in one run.

Analysis plan

Primary: at d8 and d20, seeds as registered, threshold . Secondary: Tagalog A2 versus P1.1 Table 2 and versus A0; A3; collapse binary on English. Falsification of forgetting: not material. Falsification of acquisition: Tagalog A2 not better than A0

by . Neither revives a P1.1 depth ranking.

Results

P1.1 Tagalog table is cited, not rerun. EN0, P0, and A0–A3 cells are empty until the new PDF exists and the runs complete .

Discussion rules

If is material and Tagalog A2 improved versus A0, the model learned Filipino and paid an English cost beyond extra English training. If Tagalog did not improve, it did not learn Filipino; do not call that English CF due to Filipino competence. If and Tagalog improved, sequential Tagalog at this budget did not overwrite English on BPB. P0 is what licenses the sentence “pretrained in English.”

Limitations

No EN0 numbers yet. English BPE on Tagalog text shreds Tagalog; BPB still compares bytes. WikiText-103 web English. P1.1 reconstructed split. One parent seed unless extra seeds run. Shared named entities across Wikipedias. No chat.

Conclusion

Paper 1 stays a pure Tagalog base. Paper 2 trains an English nanochat, proves provenance, continues on WikiText-TL-39, and reports English forgetting and Filipino acquisition on the same BPB instrument. That is the pipeline. It is not a rewrite of AsPredicted #306780.

Ethics

Public Wikipedia text. No passcodes, no test.jsonl in public repos. Hub license other. Cheng is not a coauthor.

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